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Virtual grid system and method

Abstract

A virtual grid system is provided that enables allocation of energy produced by an energy producing asset to individual units of a multi-unit building. The system tracks the amount of electrical energy produced by the energy producing asset and also tracks an amount of electrical energy used by each individual unit. The system compares energy production information including pairs of an amount of energy produced and a corresponding time interval during which the amount of energy was produced to energy usage information including pairs of an amount of energy used and a corresponding time interval during which the amount of energy was used. The system allocates an amount of energy produced during each time interval to the individual units based on the amount of energy consumed by each respective individual unit during the time intervals. The system also calculates an energy cost allocation for each individual unit.

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Background/Summary

RELATED APPLICATIONS (1) The present application is a National Stage entry of International Patent Application No. PCT/US2021/018894, filed on Feb. 19, 2021, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/979,919, filed on Feb. 21, 2020, the entirety of each of which is incorporated herein by reference in its entirety for all purposes.

BACKGROUND

Field of the Invention

(1) The present disclosure is generally directed toward a virtual grid system for shared use of an energy producing asset by individual units of a multi-unit residential building and more specifically directed toward systems and methods for tracking and allocating shared use of an energy producing asset by individual units of a multi-unit residential building.

Related Art

(2) Owners of multi-unit residential buildings such as apartment buildings are increasingly installing energy producing assets such as solar panels to satisfy the demand of tenants for sustainable onsite energy production and less expensive energy expenses. However, a problem with such energy producing assets is that the energy produced is partially consumed by the multi-unit residential building and various common areas and the remainder of the energy produced is pushed onto the grid. The owner of the multi-unit residential building is unable to sell the energy produced and the benefit of the energy producing asset is largely unrealized.

(3) Therefore, what is needed is a system and method that overcomes these significant problems found in the conventional systems as described above.

SUMMARY

(4) Accordingly, the present disclosure provides a virtual grid system that enables shared use of one or more distributed energy resources (“DERs”) by multi-unit buildings and allocation of the benefits of such shared use to individual units. In an embodiment, the system includes one or more DER producing assets configured to generate electrical energy and deliver the electrical energy to a multi-unit building having a plurality of individual units, each configured with a corresponding individual unit electrical panel. The system also includes a production current reader electrically coupled with the energy producing asset and configured to track an amount of electrical energy produced by the energy producing asset and communicatively coupled with a data communication

network. The system also includes a junction configured to receive electrical energy from the energy producing asset and deliver the electrical energy to a plurality of individual unit electrical panels for consumption by one or more usage appliances within the plurality of individual units. The system also includes a plurality of usage current readers, each configured to track an amount of electrical energy used by an individual unit and each communicatively coupled with the data communication network. The system also includes a virtual grid server communicatively coupled with the production current reader and the plurality of usage current readers via the data communication network. The virtual grid server is configured to obtain energy production information for the energy producing asset, wherein the energy production information comprises a plurality of pairs of an amount of energy produced and a corresponding time interval during which the amount of energy was produced. The server is also configured to obtain energy usage information for each of the individual units of the multi-unit building, wherein the energy usage information for an individual unit comprises a plurality of pairs of an amount of energy used and a corresponding time interval during which the amount of energy was used. The server is also configured to allocate a first amount of energy produced during a first time interval to each individual unit in a first plurality of individual units based on a first amount of energy consumed by each respective individual unit during the first time interval, wherein the allocated first amount of energy produced may be different for each of the individual units in the first plurality of individual units. The server is also configured to allocate a second amount of energy produced during a second time interval to each individual unit in a second plurality of individual units based on a second amount of energy consumed by each respective individual unit during the second time interval, wherein the allocated second amount of energy produced may be different for each of the individual units in the second plurality of individual units. The server is also configured to calculate an energy cost allocation for each individual unit in the multi-unit building based at least in part on the allocated first amount of energy produced during the first time interval and the allocated second amount of energy produced during the second time interval.

(5) Other features and advantages of the present invention will become more readily apparent to those of ordinary skill in the art after reviewing the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The structure and operation of the present invention will be understood from a review of the following detailed description and the accompanying drawings in which like reference numerals refer to like parts and in which:

(2) FIG. 1A is a diagram illustrating an example system for shared use of a DER producing asset according to an embodiment of the invention;

(3) FIG. 1B is a diagram illustrating an example system for shared use of a DER producing asset according to an embodiment of the invention;

(4) FIG. 1C is a diagram illustrating an example system for shared use of a DER producing asset according to an embodiment of the invention;

(5) FIG. 2 is a diagram illustrating an example system for shared use of a DER producing asset and communication system according to an embodiment of the invention;

(6) FIG. 3 is a diagram illustrating an example system for monitoring and control of energy production by a shared DER producing asset according to an embodiment of the invention;

(7) FIG. 4 is a diagram illustrating an example system for shared use of a DER producing asset according to an embodiment of the invention;

(8) FIG. 5 is a diagram illustrating an example server system for shared use of a DER producing

asset according to an embodiment of the invention;

(9) FIG. 6 is a diagram illustrating example functionality of a load allocation module according to an embodiment of the invention;

(10) FIG. 7 is a diagram illustrating example functionality of an energy bank module according to an embodiment of the invention;

(11) FIG. 8 is a diagram illustrating example functionality of a weighting module according to an embodiment of the invention;

(12) FIG. 9 is a diagram illustrating example functionality of an energy data storage module according to an embodiment of the invention;

(13) FIG. 10 is a diagram illustrating example functionality of a utility credit module according to an embodiment of the invention;

(14) FIG. 11 is a diagram illustrating example functionality of an avoided cost module according to an embodiment of the invention;

(15) FIG. 12 is a diagram illustrating example functionality of a time of use module according to an embodiment of the invention;

(16) FIG. 13 is a diagram illustrating example functionality of a water heater module according to an embodiment of the invention;

(17) FIG. 14 is a diagram illustrating example functionality of an appliance automation module according to an embodiment of the invention;

(18) FIG. 15 is a diagram illustrating example functionality of a load forecasting module according to an embodiment of the invention;

(19) FIG. 16 is a block diagram illustrating an example breakdown of total shared energy usage according to an embodiment of the invention;

(20) FIG. 17 is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention;

(21) FIG. 18 is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention;

(22) FIG. 19 is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention;

(23) FIG. 20 is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention;

(24) FIG. 21 is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention; and

(25) FIG. 22 is a block diagram illustrating an example wired or wireless system that may be used in connection with various embodiments described herein.

DETAILED DESCRIPTION

(26) Embodiments disclosed herein provide for a virtual grid system that enables distributed use of electrical energy produced by an energy producing asset and allocation of such distributed use to individual units in a multi-unit building. For example, one method disclosed herein allows for energy production by the energy producing asset to be tracked in time intervals and energy usage by individual units in the multi-unit building to be similarly tracked in time intervals and then usage of the energy produced by the energy producing asset to be allocated to the individual units. After reading this description it will become apparent to one skilled in the art how to implement the invention in various alternative embodiments and alternative applications. However, although various embodiments of the present invention will be described herein, it is understood that these embodiments are presented by way of example only, and not limitation. As such, this detailed description of various alternative embodiments should not be construed to limit the scope or breadth of the present invention as set forth in the appended claims.

(27) FIG. 1A is a diagram illustrating an example system (10) for shared use of an energy producing asset (20) by individual units in a multi-unit building (60) according to an embodiment

of the invention. In the illustrated embodiment, three solar systems (20A) (20B) (20C) are connected to three individual electrical panels (30A) (30B) (30C) associated to specific individual units. This diagram illustrates a 1:1 relationship between solar arrays and individual units. This is an inefficient and expensive method to use solar production to offset energy consumption due to the need for multiple wired connection runs between the solar system and the electrical panels.

(28) FIG. 1B is a diagram illustrating an example system (10) for shared use of an energy producing asset (20) according to an embodiment of the invention. This diagram illustrates a single solar system (20) connecting to a DC/AC junction (80) that is coupled the main building electrical bussing (40) alongside a potential battery (70). The main electrical bussing (40) is now linked to the utility grid via transformer (50), the DC/AC junction (80) and the individual dwelling unit electrical panels (30A) (30B) (30C). This system allows for the solar production and discharged storage to be shared throughout the individual dwelling units of the multifamily dwelling (60).

(29) FIG. 1C is a diagram illustrating an example system (10) for shared use of an energy producing asset (20) according to an embodiment of the invention. This diagram represents a single solar system (20) connecting to a DC/AC junction (60) that is coupled to a utility transformer (50) that feeds the building electrical bussing (40) alongside a potential battery (70). The utility transformer (50) is linked to the utility grid, the DC/AC junction (80) and the main building electrical bussing (40). The main building electrical bussing (40) feeds the individual dwelling unit electrical panels (30A) (30B) (30C), allowing the solar energy produced and the energy stored in the battery (70) that gets discharged to be shared throughout the individual dwelling units.

(30) FIG. 2 is a diagram illustrating an example physical configuration of a virtual grid server and related components in a system (10) that correlate energy data and actions within the virtual grid network according to the embodiment of the invention. In the illustrated embodiment, energy consumption is monitored through existing or new current transmitter (“CT”) Readers (100) linked to each individual dwelling unit that relay real-time usage data to a central gateway system (110) connected to the site network (120). A CT Reader (100) and a control switch (130) could be a combined component allowing electron usage control to be determined by the network (120). The energy usage nodes (140) inside the network comprise major appliances such as an electric vehicle charger or water heater, or unitary real estate within a property such as a leased unit or separate building on a property. A CT Reader (100) could comprise but is not limited to a device such as a utility meter, a magnetic current reader as a stand-alone component, or an embedded CT Reader inside of an appliance. The data collected is interpreted by the Ivy Server and used for purposes as described later with respect to FIG. 5. In one embodiment, API language can be used to access energy usage node data stores.

(31) FIG. 3 is a diagram illustrating an example system (10) for monitoring and control of energy production by a shared energy producing asset (20) according to an embodiment of the invention. Current readers (100) are used to track energy production amounts which will be connected to a gateway (110) that will send data to the network (120), also a battery Power Junction & Control Unit (150) will be accessed via the gateway (110) to allow the network (120) to control in and out electron flow to and from the battery storage unit (70).

(32) FIG. 4 is a diagram illustrating an example system (10) including electrical and data connections between elements of the system (10). In the illustrated embodiment of FIGS. 2, 3, and 4, dotted lines represent data connections and solid lines represent electrical connections.

(33) FIG. 5 is a diagram illustrating an example virtual grid server (200) and various software modules that execute on the virtual grid server (200) according to an embodiment of the invention. In the illustrated embodiment, the virtual grid server (200) includes a production module 305, a consumption module 310, an energy data storage module 315, an EV charging module 320, a storage module 325, a utility credit module 330, a load allocation module 335, an energy bank module 340, an avoided cost module 345, a water heater module 350, a time of use module 355, a forecasting module 360, a weighting module 365, a user billing module 370, an appliance

automation module **375**, an energy market module **380**, a power host module **385**, a resident user module **390**, an interface module **395**, and a reporting module **397**.

(34) The virtual grid system has a work processing design that will use triggers from these modules to perform processing tasks that accomplish the physical goals of the virtual grid system using specific modules separately. The data is collected from modules that have a data collection processing task to retrieve energy data nodes through associated API's or network connections such as the production module, consumption module, water heater module, EV charging module, and storage module. The virtual grid system then has another layer of data collection to help with the output processing used to generate real time events and value distribution outputs within the virtual grid environment such modules include the time of use module, energy market module, and utility credit module. The data collection tasks performed by these modules are different from the energy node collection work processing tasks because they are collecting real time up to date benchmarking information on external characteristics such as utility tariff changes, regional energy market pricing trends, and external pricing indications that affect the virtual benchmarks used within the virtual grid environment. Ultimately the combination of these software modules used by the virtual grid categorize these external benchmarks and the energy data nodes collected in a unique way to then process the data to creating events that drive physical value adding results within the application and enable physically applicable methodology behind the value distribution of the physically generated assets and their given value. This physical value generated is made up inside the work processing modules that each have their own dedicated purpose within the virtual grid environment.

(35) The production module **305** is configured to collect energy production data from within the physical devices within the virtual grid system and store it in the Ivy Server ready to be used by other relevant modules for their use case.

(36) The production module will store associated solar systems and inverter (CT's) energy production history and will assign the energy production assets to the correct virtual grid system so that the allocation module can use the correct datasets inside of the Ivy Server.

(37) The consumption module **310** is configured to collect the energy usage information from the different nodes within the virtual grid system and store and associate the data points to the correct physical attribute such as an appliance, real estate unit, or building.

(38) The consumption module will associate meters (CT's) to users inside of the ivy software and will API to meter management networks to refresh energy usage data streams. The consumption module will also store associated authorization data inputs needed to connect to specific meter networks such as a utility.

(39) The energy data storage module **315** is a work process module that correlates and compresses historical energy data into optimized long term storage formats.

(40) The EV charging module **320** is configured to operate similarly to the appliance module. The difference is that rather than the EV charging appliance being classified as a unit property it could be shared by users that live on the property. The EV charging module will associate users as they use the appliance to the amount of energy used and to a unit which allows the usage dataset to be formulated inside of the allocation & billing module.

(41) The storage module **325** is configured to connect to 3rd party battery load control systems and set preferences for how the Ivy Software will interface with battery management based on potential outcomes of the allocation load module. This module will then identify optimization patterns based on load balance information within the virtual grid and communicate events to the storage devices within the virtual grid system.

(42) The utility credit module **330** is configured to identify the format in which a utility credit or incentive is generated and create a work process to input the needed energy data information from the virtual grid system. The module will store preferences for making these calculations and store the different criteria or options in which the virtual grid can use.

- (43) The load allocation module **335** is configured to interpret all of the generation and usage nodes on a property and will determine the virtual generation allocation as shown and described in more detail with respect to FIG. 6.
- (44) The energy bank module **340** is configured to communicate with the allocation module to represent the price segment of surplus energy generated on a property and how that energy fits into the load allocation outputs based on time of use energy differences.
- (45) The avoided cost module **345** is configured to calculate what a unit would have paid the utility company during a given time period.
- (46) The water heater module **350** is configured to communicate load balance information from the virtual grid system to individual water heater appliances across the environment based optimal usage times or on user preferences.
- (47) The time of use module **355** is configured to be an active repository of time of use segments available to a virtual grid system based on utility pricing structures.
- (48) The forecasting module **360** is configured to use relevant data sets recorded over time inside of the ivy server to predict energy load peaks, shifts, and surplus energy generation events. The relevant data sets will be scored once real data is obtained and these scores will be used to validate future predictions based on score trends. Potential types of historical data obtained from the virtual grid system and how the data process works is shown in more detail in FIG. 15.
- (49) The weighting module **365** is configured to weight each unit's usage loads against each other based on positive usage behavior and external utility pricing signals and distribute the shared financial incentives identified by the utility credit module based on each unit's weighted usage.
- (50) The user billing module **370** is configured to pick up the outputs from the weighting and avoided cost module to generate a billing output that is delivered to a user that lives within the property. This module will reference variables such as when and how the outputs will be delivered to a user based on the settings inside of the Power Host and Resident User interface modules.
- (51) The appliance automation module **375** is configured to communicate load balance information within the virtual grid system and push event actions to those applicable appliances such as when it is optimal to use energy based on surplus generation events in real time.
- (52) Then energy market module **380** is configured to host a repository of external references related to energy market pricing that can be used and incorporated into the forecasting module and automation module to help user energy behavior adopt to energy market price indications.
- (53) The power host module **385** is configured to store preferences that indicate the desired variables associated to a property owner account or stakeholder. The variables will then reference a needed module such as equal discount logic and equal pricing information that the user billing module will use.
- (54) The resident user module **390** is configured to allow users to access their consumption data sets, virtual allocation history, avoided rate criteria and inputs, associated appliances, solar billing history, and notifications for optimizing energy behavior.
- (55) The interface module **395** is configured to allow a user with associated data inside of the virtual grid system to visualize data trends, suggestions, or preferences. Such preferences or data trends could include suggestions on when automation events would occur, what types of events they want to be notified about, and energy usage patterns.
- (56) The reporting module **397** is configured to allow owners or stakeholders involved in a property to view the different summary or detailed data sets from these modules.
- (57) FIG. 6 is a diagram illustrating an example functionality of the load allocation module (**335**) according to an embodiment of the invention. In the illustrated embodiment, the load allocation module (**335**) is configured to identify all of the usage benchmarks inside of a property behind the service delivery point. The load allocation module (**335**) is configured to automatically run during each time interval of 6 seconds-15 minutes based on a predetermined election. This mathematical process will be called a data process loop. The loop will apply the shared generation load across the

multiple usage nodes to a virtual allocation ledger associated to the user or Multiple Unit Dwelling (“MUD”) based on their real consumption at that time. This load allocation module (335) will equally distribute the generation load across ledgers up to the usage benchmark of a node equally. This output metric is called virtual distributed energy resource allocation ledger. This load allocation module (335) will also communicate with the energy bank module and determine the time stamped energy bank decrease or increase amounts within the loop according to load net differences. This load allocation module (335) will also determine a property wide aggregated need for electricity from beyond the utility electricity delivery point meaning that each interval there will either be a credit generated for too much electricity produced vs consumed or a need for electricity from outside the property.

(58) FIG. 7 is a diagram illustrating an example functionality of the energy bank module (340) according to an embodiment of the invention. In the illustrated embodiment, the energy bank module (340) is configured to keep track of surplus produced energy given real time of generation and usage comparison which will be identified in the allocation module. For example if 20 kWh of extra energy is assessed to be generated at a given time period the 20 kWh will be associated to a specific Time Of Use “TOU” bank availability to be tracked and applied at a later time. The reapplying process will first use availability from the direct TOU associated to the usage need identified in the allocation module and then will follow a priority list. If the same TOU segment has no availability it will pull from the next TOU bank in line but the credit will be referenced at the correct TOU value from when it was generated vs used so that the credit is weighted fairly in the weighting module. This data will be referenced in the load allocation module (335) and weighting module (365).

(59) FIG. 8 is a diagram illustrating an example functionality of the weighting module (365) functionality according to an embodiment of the invention. In the illustrated embodiment, process is outlined that is configured to distribute the shared financial incentives identified by the utility credit module based on unique usage behavior per user. This weighted module (365) will factor avoided time of use cost for allocated generation load as described in the allocation module. The core functionality of this weighted module (365) is weighting the unit usage loads against each other based on positive usage behavior and external utility pricing signals. A % value at the end of a billing cycle per unit will determine how much of the aggregated property utility credit will be associated to each user. The rules will be equally available to all and are based on real time energy pricing weighted by utility price reference. This weighting module (365) will update based on utility time of use updates to incorporate real time market pricing into the weighting calculation.

(60) FIG. 9 is a diagram illustrating example storage repository segments of the energy data storage module (315) according to an embodiment of the invention. In the illustrated embodiment, the energy data storage module (315) and the objects relationship structure are shown. There are four main energy data objects; 1) The entire property or service delivery area. 2) A building within the property or service area. 3) A real estate unit that has a need to be identified separately within the service area usually because of lease obligation or disaggregated ownership. 4) An appliance or usage node within a unit.

(61) FIG. 10 is a diagram illustrating an example utility credit module (330) according to an embodiment of the invention. In the illustrated embodiment, the utility credit module (330) is configured to associate energy production data with the utility incentive schema of allocated energy generation on a property and identify the utility recorded value credited for produced energy. This module (330) will calculate as the utility would the end result as a total savings or credit generated per billing time period as the utility would have on record. The dollar amounts will be allocated to the entire service area and will then be used in the weighting module to generate individual user savings figures. This module (330) will create an estimated energy cost figure known as “Net” that includes the savings from solar off of the “gross” cost that would have been due to the utility. This figure will be available before the utility publishes their final NET amount due. This verified NET

amount will be used in the weighting module and if the weighting module used the predetermined NET amount it will calculate the difference between the two and incorporate it into the next following billing cycle. This module (330) will adjust based on any net metering changes associated with physical tariff components and incentives.

(62) FIG. 11 is a diagram illustrating an example avoided cost module (345) according to an embodiment of the invention. In the illustrated embodiment, the avoided cost module (345) is configured to determine the cost a user would have paid in energy cost on a certain utility tariff, based on the associated current reader's unique energy usage. This cost total does not factor solar production and is only based on gross consumption. This module (345) includes dynamic input fields that are adjusted based on the associated tariff's variable criteria needed to 100% accurately calculate the avoided tariff, and the module (345) will store and reference the criteria preferences based on the physical answers that a resident living in the associated unit has chosen inside of an interface module.

(63) FIG. 12 is a diagram illustrating an example time of use module (355) according to an embodiment of the invention. In the illustrated embodiment, the time of use module (355) is configured to pull data points from public utility tariffs, and store them per time of use segment. Time of use segments are determined based on a utility tariff and comprise a time period with a set energy cost per that time of use. A utility tariff is a rate structure associated to a utility company and this module (355) will pull relevant tariff data and keep the repository segment up to date with live tariffs being used in the regional locations that the ivy software server is contracted to function in.

(64) FIG. 13 is a diagram illustrating an example water heater module (350) according to an embodiment of the invention. In the illustrated embodiment, the water heater module (350) is configured to use interval load outputs from the load allocation module (335) to determine when is the optimal time to heat water on an automated schedule based on surplus energy being generated on the property.

(65) FIG. 14 is a diagram illustrating an example appliance automation module (375) according to an embodiment of the invention. In the illustrated embodiment, the appliance automation module (375) is configured to segment physical environments by property. This module (375) will run live event protocol to load control switches to turn off or reduce energy usage, battery management systems to increase stored current or discharge stored energy capacity, or user interfaces to prompt energy usage behavior change based on the following; Surplus onsite energy available within the physical environment as described in the load allocation module (335), utility or outside of physical environment pricing incentives to reduce energy usage such as the time of use module reference.

(66) FIG. 15 is a diagram illustrating an example load forecasting module (360) according to an embodiment of the invention. In the illustrated embodiment, the components used inside of the load forecasting module (360) are shown. This module (360) is configured to create prediction tracks that consist of a data formulation process that will have similar data points referenced in the figure below and is meant to associate energy usage and shift changes across energy. The goal of this module is to produce patterns that can be layered on top of live and other future data sets to predict and test prediction load usage and shifts in usage in a scoring system that is based on verified accuracy. This module (360) with a niche focus of optimizing energy usage behavior in the physical environment is described in FIG. 1.

(67) FIG. 16 is a block diagram illustrating an example breakdown of total shared energy usage according to an embodiment of the invention. In the illustrated embodiment, a plurality of users have a total shared energy usage 1605 during a time period, for example a billing cycle. The users share energy produced by one or more DER producing assets supplemented by energy from the grid managed by the utility. The breakdown of the cost of the total shared energy usage 1605 is divided between a non-avoided utility energy cost 1610 and an avoided utility energy cost 1615. Payment is made to the utility to cover the non-avoided utility energy cost 1610 and payment is

made to the DER Host to cover at least a portion of the avoided utility energy cost **1615**. FIGS. **17-21** will now be described to discuss alternative methods for determining how to allocate the shared benefit of the one or more DER producing assets to arrive at payment that is made to the DER host to cover at least a portion of the avoided utility energy cost **1615**.

(68) In accordance with the description of FIGS. **17-21**, it should be noted that these methods may be carried out by a server system **200** such as previously described with respect to FIG. **5** and that the various steps of the methods illustrated in FIGS. **17-21** may be carried out by various modules of the server system **200**.

(69) FIG. **17** is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention. Initially at **1620** the server determines a fixed % of DER generation credits per unit in the multi-unit building. Next, at **1622** the system calculates a DER payment per unit using the fixed % of generation credits allocated per unit. For example, if there are 100 units, each unit may be allocated 1% of the DER generation credits and the DER payment is reduced by the value of 1% of the DER generation credits at **1622**. Next, the tenant for each unit remits DER energy payments to the DER host at **1624** and the tenant for each unit also remits energy payments to the utility at **1626**, although these last two steps can be in any order.

(70) In one example embodiment, the DER generation credits are allocated to each unit using a fixed percentage. Then a fixed value per kWh is used to calculate the amount of DER generation kWh credits for each unit to determine the payment a tenant of a unit makes to the DER host. The tenant of each unit then makes a payment to the DER host and also to the utility.

(71) FIG. **18** is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention. Initially, at **1630** the server obtains energy use data for each unit. Then at **1632** the server calculates the gross utility cost for the aggregate energy used by the multi-unit building for the time period, e.g., a billing cycle. The gross utility cost is the total energy used during the time period at the cost the utility would charge for that total energy used. Next, at **1634** the server determines a discount as a percentage or a flat dollar amount. The discount is based on the benefit of using energy from the one or more DER producing assets during the time period rather than getting that same energy from the utility. The server also applies the discount to the gross utility cost for each unit to reduce the cost of energy to each unit. Next, at **1636**, the tenant for each unit remits a DER energy payment to the DER host and at **1638**, the DER host remits energy payments to the utility, although these last two steps can be in any order.

(72) In one example embodiment, energy usage data is collected at a residential unit level normally from a utility meter or current transformer in 15-60 minute time intervals. Then a reference is made to the public utility tariff associated to each tenant's billing profile and the Gross Utility Cost is calculated to determine the energy cost for each residential unit without the DER benefit. The server determines a flat percentage reduction or a fixed dollar amount to be deducted from the gross utility cost of a unit to determine the energy payment amount for each tenant. Then each tenant pays the DER host and the host in turn pays the utility.

(73) FIG. **19** is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention. Initially at **1650**, the server obtains energy use data for each unit in the multi-unit building. Next at **1652**, the server calculates the gross energy cost per unit. The gross energy cost for a unit is the amount that the utility would have charged the unit for energy use by the unit during the timer period, e.g., one billing cycle. Next, at **1654**, the server determines the DER benefit per unit. In one embodiment, the DER benefit for a unit may be based on the relative energy use of that unit compared to all other units that share use of the one or more DER producing assets.

(74) Next, at **1656** the server determines the aggregate DER benefits and then at **1658** the server reduces the gross energy cost per unit by a fixed percentage of the aggregate DER benefits previously determined. Subsequently, at **1660**, the tenant for each unit remits a DER energy

payment to the DER host and at **1662**, the DER host remits energy payments to the utility, although these last two steps can be in any order.

(75) In one example embodiment, energy usage data is collected at a residential unit level, for example, from a utility meter or current transformer in 15-60 minute time intervals during the time period. Then a reference is made to the public utility tariff associated to each tenant's billing profile and the Gross Utility Cost is calculated per unit without the DER benefits. The Gross Energy cost for a unit during the time period is then compared to Net Energy cost for the same unit, where the net energy cost is determined by reducing the gross energy cost based on the DER benefit for the unit. The difference between the gross energy cost and the net energy cost is the "Avoided Energy Cost". The sum of all the units avoided energy cost will be totaled into a total community energy savings. A fixed % of these total community energy savings are then deducted from the Gross Energy cost associated to each residential unit and then the tenant makes a corresponding DER payment to the DER host and the DER host makes a corresponding payment to the utility.

(76) FIG. **20** is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention. Initially at **1670**, the server obtains energy use data for each unit in the multi-unit building. Next at **1672**, the server calculates the gross energy cost per unit. The gross energy cost for a unit is the amount that the utility would have charged the unit for energy use by the unit during the time interval, e.g., one billing cycle. Next, at **1674**, the server determines the DER benefit index per unit. In one embodiment, the DER benefit index for a unit may be based on the relative energy use of that unit compared to all other units that share use of the one or more DER producing assets.

(77) Next, at **1676** the server determines the aggregate DER benefits and then at **1678** the server determines the DER benefits index for each unit. There are a variety of ways to determine the DER benefits index for a unit and one such method is described with respect to FIG. **21**. Next, at **1680**, the server reduces the gross energy cost per unit in accordance with the DER benefits index for the respective unit and generates the DER energy payment amount for each unit. Subsequently, at **1682** the tenant for each unit remits a DER energy payment to the DER host and at **1684**, the DER host remits energy payment to the utility, although these last two steps can be in any order.

(78) In one example embodiment, energy usage data is collected at a residential unit level, for example from a utility meter or current transformer in 15-60 minute time intervals during the time period such as a billing cycle. Then a reference is made to the public utility tariff associated to the tenant's billing profile and the Gross Utility Cost is determined per unit without the DER benefit. The gross utility cost is then compared to a Net Utility cost that is determined based on the gross utility cost with the DER benefit. The net utility cost amount will determine the savings generated from the DER otherwise known as "Avoided Energy Cost". The sum of all avoided energy costs across all units is determined as the total community energy savings and each unit's relative share of the total community energy savings is applied to the unit's gross utility cost to determine the DER payment for each unit. This DER payment is then made by the tenant of each unit and the DER host makes a payment to the utility.

(79) FIG. **21** is a flow diagram illustrating an example process for allocating shared use DER benefits according to an embodiment of the invention. Initially at **1702** the server obtains energy use data for each unit in the multi-unit building during a time period. A time period can be for example a billing cycle established by the utility. Other time periods may also be employed. The time period is divided into a plurality of intervals. For example, the time period may be divided into 60 minute intervals or it may be divided into 15 minute intervals. The granularity of an individual interval can be very small, for example 30 seconds or even 1 second. In one embodiment, the server divides the time period into time intervals based on pricing time intervals established by the utility. For example, the utility may adjust kWh pricing every 60 minutes or every 120 minutes or every 15 minutes.

(80) Next, at **1704** the server obtains DER production data from the one or more shared DER

production assets. The DER production data may be originally produced in desired time intervals or the server may concat and/or divide the DER production data into the desired time interval, for example, the time interval that was used for the energy use data. For example, if the energy use data for a time period was divided into time intervals of 15 minutes, the DER production data is similarly divided into time intervals of 15 minutes.

(81) Next, at **1706** the server obtains kWh pricing information for the utility for each time interval during the time period. Accordingly, if the previously established time interval was 15 minutes, then the server obtains the kWh price of energy from the utility for each time interval during the time period.

(82) Next, at **1708**, the server obtains the energy use data for each unit during a first time interval of the time period. For example, if the time period is 30 days and the time interval is 15 minutes, then there are 96 time intervals per day and 2,880 time intervals for the time period. At **1708**, the server obtains the energy use data for each unit during time interval 1 of 2,880.

(83) Next, at **1710** the server determines if there was DER kWh available during the first time interval. If the one or more shared DER production assets were producing energy during the time interval, then there is some amount of DER kWh available during the time period and at **1712**, the server applies the available DER kWh to each unit. The application of available kWh to each unit may be carried out in a variety of ways. In one embodiment, kWh are applied equally to each unit up to 100% of a unit's energy usage during the time period. For example, if there were 10 units and 50 available DER kWh and each unit used 5 kWh during the time interval, then each unit would have 5 kWh of the available DER kWh applied. Alternatively, if there were 10 units and 50 available DER kWh and each unit used 10 kWh during the time interval, then each unit would have 5 kWh of the available DER kWh applied and each unit would have 5 kWh remaining during the time interval. Alternatively, if there were 10 units and 50 available DER kWh and 5 units used 5 kWh during the time interval and 5 units used 10 kWh during the time interval, then each unit would have 5 kWh of the available DER kWh applied and 5 of the units would have 5 kWh remaining during the interval.

(84) Next, after applying the available DER kWh to all of the units, at **1714** the server updates the DER benefits index per unit based on the DER kWh benefit that was applied to each individual unit for the particular time period and the utility price of a kWh of energy for the particular time interval. In one embodiment, the DER benefits index is calculated by summing the price of each quantity of DER kWh applied to the unit by the server and the price of each quantity of virtual bank kWh applied to the unit by the server during the time period.

(85) Also after applying the available DER kWh to all of the units, at **1716** the server determines if there is excess DER kWh during the time interval. It should be noted that it is unlikely that the DER kWh production and the total energy use across all units is identical, however if they are identical, then the process jumps to **1732** to determine if there are more time intervals during the time period and if there are, the process begins again at **1708** with the next time interval during the time period. In the case where there is no excess DER kWh as determined at **1716**, and the production and use was not identical so there are additional kWh used by the units during the time interval, the process jumps forward to **1724**, which will be explained below.

(86) At **1718**, after the server determines that there is excess DER kWh produced during the time interval, the server adds the excess DER kWh to a virtual bank for the time interval. The virtual bank is maintained by the server to account for variations in energy production by the one or more shared DER production assets and overall usage by the various units in the multi-unit building. The server also maintains average utility pricing information for each time interval. Importantly, in some instances, each individual unit may have a different kWh price for energy from the utility during the same time period due to complexities with tariffs and other energy pricing laws and regulations. Accordingly, for each time interval, the server maintains a virtual bank of excess kWh produced by the shared DER production assets and for each time interval, the server maintains a

quantity of kWh and an average price per kWh. For example, if the server is using 96 time intervals (i.e., 15 minute intervals), then there are 96 discrete virtual banks.

(87) At **1720**, after determining how much excess kWh will be added to the virtual bank for the particular time interval, the server determines the average utility price per kWh for the particular time interval and then at **1722**, the server updates the quantity of kWh in the virtual bank for the time interval and the also updates the average price of a kWh in the virtual bank for the time interval. Because there was excess DER kWh and after the excess DER kWh are added to the virtual bank, the process jumps forward to **1732** to determine if there are more time intervals during the time period and if there are, the process begins again at **1708** with the next time interval during the time period.

(88) If the server determines that there is no DER kWh available for the particular time interval at **1710** or the server determines at **1716** that there is no excess DER kWh (and the available DER kWh is not identical to the total energy use during the time interval), the server determines at **1724** if there is any virtual bank kWh available. If there is no kWh available in the virtual bank, the process jumps to **1732** to determine if there are more time intervals during the time period and if there are, the process begins again at **1708** with the next time interval during the time period.

(89) If the server determines at **1724** that there is kWh available in the virtual bank, at **1726** the server applies available kWh in the virtual bank to each unit. The application of available kWh from the virtual bank to each unit may be carried out in a variety of ways. In one embodiment, available kWh can be applied equally to each unit up to 100% of a unit's energy usage during the time period as previously described.

(90) In one embodiment, a difference between application of DER kWh to the units and application of virtual bank kWh is the time interval. For example, when the DER kWh are being applied, the available DER kWh is only from the current time interval. In contrast, when virtual bank kWh are being applied, the kWh may come from the current time interval or a different time interval. In one embodiment, the server is configured to first select available virtual bank kWh from the current time interval and if there is no available virtual bank kWh from the current time interval, then the server is configured to select available virtual bank kWh from the current time interval on a different day during the time period, and if there is no available virtual bank kWh from the current time interval on a different day during the time period, then the server is configured to select the lowest price kWh that is available from the virtual bank during the time period.

(91) In an alternative embodiment, the server is configured to first select available virtual bank kWh from the current time interval and if there is no available virtual bank kWh from the current time interval, then the server is configured to select available virtual bank kWh from the current time interval on a different day during the time period, and if there is no available virtual bank kWh from the current time interval on a different day during the time period, then the server is configured to select the closest price kWh that is available from the virtual bank during the time period.

(92) Next, at **1728**, the server updates the quantity of kWh in the virtual bank for each time interval from which available kWh were applied and the server the also updates the average price of a kWh in the virtual bank for each time interval from which available kWh were applied.

(93) Next, at **1730**, the server updates the DER benefits index for each unit that received available kWh from the virtual bank based on the quantity of kWh applied from the virtual bank and the average price of each kWh of energy applied to the unit. Advantageously, the virtual bank maintains for each time interval during the time period, a quantify of available kWh and an average price of a kWh in each time interval. In one embodiment, the DER benefits index is calculated by summing the price of each quantity of DER kWh applied to the unit by the server and the price of each quantity of virtual bank kWh applied to the unit by the server during the time period.

(94) Next, at **1732** the server determines if there are more time intervals during the time period and if there are, the process begins again at **1708** with the next time interval during the time period. If

there are no additional time intervals for the time period, at **1734** the server begins the process of generating individual energy bills having a DER payment amount for each unit.

(95) In one example embodiment, a full billing time period of energy usage data is collected across all time intervals during the time period, where the time interval can be of any duration between 30 second intervals and 60 minute intervals. After collection of the energy use data, the benefit of energy production by the shared one or more DER production assets is applied to the various units. Initially, the server determines the total energy use for each unit for each time interval during the time period. Then the server determines if DER kWh production was available during the time interval and if so, how much. If DER kWh was available, the kWh is distributed equally to every unit until the unit's total energy use during the time interval is met 100% with DER energy production. If one unit's energy use is fully met during a time interval, the server continues to equally apply the DER kWh to any remaining units with additional kWh usage until the energy usage of all units has been 100% met or until the DER kWh production has been 100% allocated.

(96) Once the DER kWh allocations are made to the units, the kWh value is referenced to an associated utility rate schedule and a contribution is made to the virtual price sum for the billing time period. If all energy usage is met and there is still remaining DER kWh production for that time period, the remaining DER kWh are contributed to the particular time interval repository in a virtual energy bank maintained by the server. When contributing DER kWh to the virtual bank, the server identified the kWh price for all units and determines an average cost of energy per kWh for the time interval across the entire community and then makes the virtual energy bank contribution with the attributes of a quantify of kWh and an average cost per kWh for the particular time interval.

(97) If all of the DER kWh production during the time interval is allocated to units and the usage is not 100% met across all units, then the server makes deductions from the virtual energy bank and applies additional kWh from the virtual energy bank to the units where usage during the time interval is not 100% met. The server is configured to apply available kWh from the virtual bank in the following order: 1) the current time interval on any other day during the time period; and 2) the lowest price per kWh in the virtual bank until. Available kWh hours from the virtual bank are applied to the energy use of any units with remaining energy use until the bank is empty or all unit usage has been 100% met. As allocations of kWh from the various time intervals in the virtual bank are applied to units, the DER benefit index of the unit is updated based on the quantity of kWh's applied and the corresponding average kWh price of the kWh applied from the virtual bank.

(98) The server follows the same process for all time intervals in the billing time period and once the application of all DER kWh and virtual bank kWh is completed, the server finalizes the DER benefit index for each unit for the particular time period, after which the server is configured to determine the final DER payment amount for each unit for the time period.

(99) FIG. **22** is a block diagram illustrating an example wired or wireless system **2200** that may be used in connection with various embodiments described herein. For example, system **2200** may be used as or in conjunction with one or more of the functions, processes, or methods (e.g., to store and/or execute the application or one or more software modules of the application) described herein, and may represent components of the virtual grid server, utility server, gateway, control switch, current reader, energy usage node, and the like as previously described and/or other processing devices described herein. System **2200** can be a server or any conventional personal computer, or any other processor-enabled device that is capable of wired or wireless data communication. Other computer systems and/or architectures may be also used, as will be clear to those skilled in the art.

(100) System **2200** preferably includes one or more processors, such as processor **2210**. Additional processors may be provided, such as an auxiliary processor to manage input/output, an auxiliary processor to perform floating-point mathematical operations, a special-purpose microprocessor having an architecture suitable for fast execution of signal-processing algorithms (e.g., digital-

signal processor), a slave processor subordinate to the main processing system (e.g., back-end processor), an additional microprocessor or controller for dual or multiple processor systems, and/or a coprocessor. Such auxiliary processors may be discrete processors or may be integrated with processor **2210**. Examples of processors which may be used with system **2200** include, without limitation, the Pentium® processor, Core i7® processor, and Xeon® processor, all of which are available from Intel Corporation of Santa Clara, California.

(101) Processor **2210** is preferably connected to a communication bus **2205**. Communication bus **2205** may include a data channel for facilitating information transfer between storage and other peripheral components of system **2200**. Furthermore, communication bus **2205** may provide a set of signals used for communication with processor **2210**, including a data bus, address bus, and/or control bus (not shown). Communication bus **2205** may comprise any standard or non-standard bus architecture such as, for example, bus architectures compliant with industry standard architecture (ISA), extended industry standard architecture (EISA), Micro Channel Architecture (MCA), peripheral component interconnect (PCI) local bus, standards promulgated by the Institute of Electrical and Electronics Engineers (IEEE) including IEEE 488 general-purpose interface bus (GPIB), IEEE 696/S-100, and/or the like.

(102) System **2200** preferably includes a main memory **2215** and may also include a secondary memory **2220**. Main memory **2215** provides storage of instructions and data for programs executing on processor **2210**, such as one or more of the functions and/or modules discussed herein. It should be understood that programs stored in the memory and executed by processor **2210** may be written and/or compiled according to any suitable language, including without limitation C/C++, Java, JavaScript, Perl, Visual Basic, .NET, and the like. Main memory **2215** is typically semiconductor-based memory such as dynamic random access memory (DRAM) and/or static random access memory (SRAM). Other semiconductor-based memory types include, for example, synchronous dynamic random access memory (SDRAM), Rambus dynamic random access memory (RDRAM), ferroelectric random access memory (FRAM), and the like, including read only memory (ROM).

(103) Secondary memory **2220** may optionally include an internal medium **2225** and/or a removable medium **2230**. Removable medium **2230** is read from and/or written to in any well-known manner. Removable storage medium **2230** may be, for example, a magnetic tape drive, a compact disc (CD) drive, a digital versatile disc (DVD) drive, other optical drive, a flash memory drive, and/or the like.

(104) Secondary memory **2220** is a non-transitory computer-readable medium having computer-executable code (e.g., disclosed software modules) and/or other data stored thereon. The computer software or data stored on secondary memory **2220** is read into main memory **2215** for execution by processor **2210**.

(105) In alternative embodiments, secondary memory **2220** may include other similar means for allowing computer programs or other data or instructions to be loaded into system **2200**. Such means may include, for example, a communication interface **2240**, which allows software and data to be transferred from external storage medium **2245** to system **2200**. Examples of external storage medium **2245** may include an external hard disk drive, an external optical drive, an external magneto-optical drive, and/or the like. Other examples of secondary memory **2220** may include semiconductor-based memory, such as programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable read-only memory (EEPROM), and flash memory (block-oriented memory similar to EEPROM).

(106) As mentioned above, system **2200** may include a communication interface **2240**. Communication interface **2240** allows software and data to be transferred between system **2200** and external devices (e.g. printers), networks, or other information sources. For example, computer software or executable code may be transferred to system **2200** from a network server (e.g., virtual grid server **200**) via communication interface **2240**. Examples of communication interface **2240**

include a built-in network adapter, network interface card (NIC), Personal Computer Memory Card International Association (PCMCIA) network card, card bus network adapter, wireless network adapter, Universal Serial Bus (USB) network adapter, modem, a wireless data card, a communications port, an infrared interface, an IEEE 1394 fire-wire, and any other device capable of interfacing system **2200** with a network (e.g., network(s) **120**) or another computing device. Communication interface **2240** preferably implements industry-promulgated protocol standards, such as Ethernet IEEE 802 standards, Fiber Channel, digital subscriber line (DSL), asynchronous digital subscriber line (ADSL), frame relay, asynchronous transfer mode (ATM), integrated digital services network (ISDN), personal communications services (PCS), transmission control protocol/Internet protocol (TCP/IP), serial line Internet protocol/point to point protocol (SLIP/PPP), and so on, but may also implement customized or non-standard interface protocols as well.

(107) Software and data transferred via communication interface **2240** are generally in the form of electrical communication signals **2255**. These signals **2255** may be provided to communication interface **2240** via a communication channel **2250**. In an embodiment, communication channel **2250** may be a wired or wireless network (e.g., network(s) **120**), or any variety of other communication links. Communication channel **2250** carries signals **2255** and can be implemented using a variety of wired or wireless communication means including wire or cable, fiber optics, conventional phone line, cellular phone link, wireless data communication link, radio frequency (“RF”) link, or infrared link, just to name a few.

(108) Computer-executable code (e.g., computer programs, such as the disclosed application, or software modules) is stored in main memory **2215** and/or secondary memory **2220**. Computer programs can also be received via communication interface **2240** and stored in main memory **2215** and/or secondary memory **2220**. Such computer programs, when executed, enable system **2200** to perform the various functions of the disclosed embodiments as described elsewhere herein.

(109) In this description, the term “computer-readable medium” is used to refer to any non-transitory computer-readable storage media used to provide computer-executable code and/or other data to or within system **2200**. Examples of such media include main memory **2215**, secondary memory **2220** (including internal memory **2225**, removable medium **2230**, and external storage medium **2245**), and any peripheral device communicatively coupled with communication interface **2240** (including a network information server or other network device). These non-transitory computer-readable media are means for providing executable code, programming instructions, software, and/or other data to system **2200**.

(110) In an embodiment that is implemented using software, the software may be stored on a computer-readable medium and loaded into system **2200** by way of removable medium **2230**, I/O interface **2235**, or communication interface **2240**. In such an embodiment, the software is loaded into system **2200** in the form of electrical communication signals **2255**. The software, when executed by processor **2210**, preferably causes processor **2210** to perform one or more of the processes and functions described elsewhere herein.

(111) In an embodiment, I/O interface **2235** provides an interface between one or more components of system **2200** and one or more input and/or output devices. Example input devices include, without limitation, sensors, keyboards, touch screens or other touch-sensitive devices, biometric sensing devices, computer mice, trackballs, pen-based pointing devices, and/or the like. Examples of output devices include, without limitation, other processing devices, cathode ray tubes (CRTs), plasma displays, light-emitting diode (LED) displays, liquid crystal displays (LCDs), printers, vacuum fluorescent displays (VFDs), surface-conduction electron-emitter displays (SEDs), field emission displays (FEDs), and/or the like. In some cases, an input and output device may be combined, such as in the case of a touch panel display (e.g., in a smartphone, tablet, or other mobile device).

(112) System **2200** may also include optional wireless communication components that facilitate

wireless communication over a voice network and/or a data network (e.g., gateway system(s) **110**). The wireless communication components comprise an antenna system **2270**, a radio system **2265**, and a baseband system **2260**. In system **2200**, radio frequency (RF) signals are transmitted and received over the air by antenna system **2270** under the management of radio system **2265**.

(113) In an embodiment, antenna system **2270** may comprise one or more antennae and one or more multiplexors (not shown) that perform a switching function to provide antenna system **2270** with transmit and receive signal paths. In the receive path, received RF signals can be coupled from a multiplexor to a low noise amplifier (not shown) that amplifies the received RF signal and sends the amplified signal to radio system **2265**.

(114) In an alternative embodiment, radio system **2265** may comprise one or more radios that are configured to communicate over various frequencies. In an embodiment, radio system **2265** may combine a demodulator (not shown) and modulator (not shown) in one integrated circuit (IC). The demodulator and modulator can also be separate components. In the incoming path, the demodulator strips away the RF carrier signal leaving a baseband receive audio signal, which is sent from radio system **2265** to baseband system **2260**.

(115) If the received signal contains audio information, then baseband system **2260** decodes the signal and converts it to an analog signal. Then the signal is amplified and sent to a speaker. Baseband system **2260** also receives analog audio signals from a microphone. These analog audio signals are converted to digital signals and encoded by baseband system **2260**. Baseband system **2260** also encodes the digital signals for transmission and generates a baseband transmit audio signal that is routed to the modulator portion of radio system **2265**. The modulator mixes the baseband transmit audio signal with an RF carrier signal, generating an RF transmit signal that is routed to antenna system **2270** and may pass through a power amplifier (not shown). The power amplifier amplifies the RF transmit signal and routes it to antenna system **2270**, where the signal is switched to the antenna port for transmission.

(116) Baseband system **2260** is also communicatively coupled with processor **2210**, which may be a central processing unit (CPU). Processor **2210** has access to data storage areas **2215** and **2220**. Processor **2210** is preferably configured to execute instructions (i.e., computer programs, such as the disclosed application, or software modules) that can be stored in main memory **2215** or secondary memory **2220**. Computer programs can also be received from baseband processor **2260** and stored in main memory **2210** or in secondary memory **2220**, or executed upon receipt. Such computer programs, when executed, enable system **2200** to perform the various functions of the disclosed embodiments.

(117) It should be understood that the previously described processes may be embodied in one or more software modules that are executed by one or more hardware processors (e.g., processor **2210**), for example, the server modules discussed herein (e.g., server modules **305-397**), which may be executed wholly by processor(s) of the virtual grid server, utility server, gateway, control switch, current reader, energy usage node, and the like as previously described, or may be distributed across these processors such that some portions or modules of the application are executed by the virtual grid server and other portions or modules of the application are executed by other processor(s) on other devices. The described processes may be implemented as instructions represented in source code, object code, and/or machine code. These instructions may be executed directly by hardware processor(s) **2210**, or alternatively, may be executed by a virtual machine operating between the object code and hardware processors **2210**. In addition, the disclosed application may be built upon or interfaced with one or more existing systems.

(118) Alternatively, the described processes may be implemented as a hardware component (e.g., general-purpose processor, integrated circuit (IC), application-specific integrated circuit (ASIC), digital signal processor (DSP), field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, etc.), combination of hardware components, or combination of hardware and software components. To clearly illustrate the interchangeability of hardware and

software, various illustrative components, blocks, modules, circuits, and steps are described herein generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled persons can implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the invention. In addition, the grouping of functions within a component, block, module, circuit, or step is for ease of description. Specific functions or steps can be moved from one component, block, module, circuit, or step to another without departing from the invention.

(119) Furthermore, while the processes, described herein, are illustrated with a certain arrangement and ordering of subprocesses, each process may be implemented with fewer, more, or different subprocesses and a different arrangement and/or ordering of subprocesses. In addition, it should be understood that any subprocess, which does not depend on the completion of another subprocess, may be executed before, after, or in parallel with that other independent subprocess, even if the subprocesses are described or illustrated in a particular order.

(120) The above description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the invention. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles described herein can be applied to other embodiments without departing from the spirit or scope of the invention. Thus, it is to be understood that the description and drawings presented herein represent a presently preferred embodiment of the invention and are therefore representative of the subject matter which is broadly contemplated by the present invention. It is further understood that the scope of the present invention fully encompasses other embodiments that may become obvious to those skilled in the art and that the scope of the present invention is accordingly not limited.

Claims

1. A method of operating a virtual grid system comprising one or more processors, the method comprising: for a plurality of units in a multi-unit building, each unit being electrically connected to at least one shared energy producing asset generating electrical energy, each unit including a corresponding unit electrical panel, the energy being delivered to respective unit electrical panels for consumption by one or more usage appliances within each of the plurality of units, obtaining energy use information, the energy use information corresponding to a time period having a plurality of time intervals, the shared energy producing asset being co-located with the plurality of units and separated from a utility grid via a utility transformer; obtaining energy production information for the energy producing asset from a production current reader via a virtual grid server communicatively coupled with the production current reader via a data communication network, the production current reader being electrically coupled with the energy producing asset, the production current reader tracking an amount of electrical energy produced by the energy producing asset, the production current reader being communicatively coupled with the data communication network, the energy producing asset providing electrical energy to a junction that delivers the electrical energy to the respective unit electrical panels for consumption by one or more usage appliances within the plurality of units, the energy production information corresponding to the time period and the plurality of time intervals; obtaining energy pricing information for each unit in the plurality of units, the energy pricing information corresponding to the time period and the plurality of time intervals, each unit having a per unit per time interval price of energy and wherein a first per unit per time interval price of energy for a first unit during a first time interval is not equal to a second per unit per time interval price of energy for a second unit during the first time interval; and for each time interval in the time period: identifying a quantity of energy used by each unit during the time interval; identifying a quantity of energy produced by the

energy producing asset during the time interval; applying at least a portion of the quantity of energy produced during the time interval to each unit's respective energy used during the time interval; if there is an excess of energy produced during the time interval after application of energy produced to each unit, adding the excess of energy produced during the time interval to a shared virtual bank for the time interval; if there is an excess of energy used by each unit during the time interval, applying at least a portion of energy from a shared virtual bank for any time interval to each unit's respective energy used during the time interval; after application of energy produced during the time interval and energy from a shared virtual bank for any time interval to each unit's respective energy used during the time interval, identifying any excess quantity of energy used by each respective unit during the time interval; and calculating an energy use cost allocation for each respective unit based on: the quantity of energy produced by the shared energy producing asset applied to the respective unit during each time interval; the quantity of energy from one or more shared virtual banks applied to the respective unit during each time interval; the excess quantity of energy used by the respective unit during each time interval; and the energy pricing information for each time interval.

2. The method of claim 1, further comprising, after applying at least a portion of the quantity of energy produced during the time interval to each unit's respective energy used during the time interval: determining an average price of energy during the time interval, the average price based on a per unit price of energy during the time interval for least a portion of the plurality of units; and updating an index for each unit based on the quantity of energy applied to the respective unit for the time interval and the determined average price of energy during the time interval.

3. The method of claim 1, further comprising, after applying at least a portion of energy from the shared virtual bank to each unit's respective energy used during the time interval: identifying one or more shared virtual bank time intervals from which energy was applied to a respective unit's energy used during the time interval; identifying an average price of the quantity of energy in each of the one or more shared virtual bank time intervals; and updating an index for each unit based on the quantity of energy from each of the one or more shared virtual bank time intervals applied to the respective unit for the time interval and the average price of the quantity of energy in each of the one or more shared virtual bank time intervals.

4. The method of claim 1, wherein the energy use information is obtained from a utility company.

5. The method of claim 1, wherein the energy use information for a particular unit is obtained from an energy meter for the particular unit.

6. The method of claim 1, wherein: the energy use information for a particular unit is obtained from one or more usage current readers; and each usage current reader is configured to track an amount of electrical energy used.

7. The method of claim 1, wherein the energy production information is obtained from one or more production current readers electrically coupled with the energy producing asset and configured to track an amount of electrical energy produced by the energy producing asset.

8. A method of operating a virtual grid system comprising one or more processors, the method comprising: for a plurality of units in a multi-unit building, each unit being electrically connected to at least one shared energy producing asset generating electrical energy, each unit including a corresponding unit electrical panel, the energy being delivered to respective unit electrical panels for consumption by one or more usage appliances within each of the plurality of units, obtaining energy use information, the energy use information corresponding to a time period having a plurality of time intervals, the shared energy producing asset being co-located with the plurality of units and separated from a utility grid via a utility transformer; obtaining energy production information for the energy producing asset from a production current reader via a virtual grid server communicatively coupled with the production current reader via a data communication network, the production current reader being electrically coupled with the energy producing asset, the production current reader tracking an amount of electrical energy produced by the energy

producing asset, the production current reader being communicatively coupled with the data communication network, the energy producing asset providing electrical energy to a junction that delivers the electrical energy to the respective unit electrical panels for consumption by one or more usage appliances within the plurality of units, the energy production information corresponding to the time period and the plurality of time intervals; obtaining energy pricing information for each unit in the plurality of units, the energy pricing information corresponding to the time period and the plurality of time intervals, wherein each unit has a per unit per time interval price of energy and wherein a first per unit per time interval price of energy for a first unit during a first time interval is not equal to a second per unit per time interval price of energy for a second unit during the first time interval; and for each time interval in the time period: identifying a quantity of energy used by each unit during the time interval; identifying a quantity of energy produced by the energy producing asset during the time interval; identifying a quantity of excess energy produced during the time interval; determining an average price of energy during the time interval, the average price based on the per unit per time interval price of energy during the time interval for at least a portion of the plurality of units; and contributing the quantity of excess energy produced during the time interval to a shared virtual bank for the time interval, where the contribution comprises the quantity of excess energy produced during the time interval and the determined average price of energy during the time interval.

9. The method of claim 8, wherein the per unit per time interval price of energy during the first time interval is determined by a utility company.

10. A virtual grid system, comprising: at least one energy producing asset configured to generate electrical energy and deliver the electrical energy to a multi-unit building having a plurality of individual units, each unit being configured with a corresponding individual unit electrical panel, the shared energy producing asset being co-located with the plurality of units and separated from the utility grid via a utility transformer; a production current reader electrically coupled with the energy producing asset, the production current reader being configured to track an amount of electrical energy produced by the energy producing asset, the production current reader being communicatively coupled with a data communication network; a junction configured to: receive electrical energy from the energy producing asset; and deliver the electrical energy to the plurality of individual unit electrical panels for consumption by one or more usage appliances within the plurality of individual units; a plurality of usage current readers, each of the plurality of usage current readers being configured to track an amount of electrical energy used by an individual unit, each of the plurality of usage current readers being communicatively coupled with the data communication network; and a virtual grid server communicatively coupled with the production current reader and the plurality of usage current readers via the data communication network, the virtual grid server being configured to: obtain energy production information for the energy producing asset, the energy production information comprising: a quantity of energy produced; and a corresponding time interval during which the quantity of energy was produced; obtain energy usage information for each of the individual units of the multi-unit building, the energy usage information for an individual unit comprising: a quantity of energy used; and a corresponding time interval during which the quantity of energy was used; allocate at least a first amount of energy produced during a first time interval to each individual unit in a first plurality of individual units based on a first amount of energy consumed by each respective individual unit during the first time interval, the allocated first amount of energy produced being the same or different for each of the individual units in the first plurality of individual units; allocate at least a second amount of energy produced during a second time interval to each individual unit in a second plurality of individual units based on a second amount of energy consumed by each respective individual unit during the second time interval, the allocated second amount of energy produced being the same or different for each of the individual units in the second plurality of individual units; and calculate an energy cost allocation for each individual unit in the multi-unit building based at least in part on: the

allocated first amount of energy produced during the first time interval; and the allocated second amount of energy produced during the second time interval.

11. A virtual grid system, comprising: at least one energy producing asset configured to generate electrical energy and deliver the electrical energy to a multi-unit building having a plurality of individual units, each unit being configured with a corresponding individual unit electrical panel, the energy producing asset being co-located with the plurality of units and separated from a utility grid via a utility transformer; a production current reader electrically coupled with the energy producing asset, the production current reader being configured to track an amount of electrical energy produced by the energy producing asset, the production current reader being communicatively coupled with a data communication network; a junction configured to: receive electrical energy from the energy producing asset; and deliver the electrical energy to the plurality of individual unit electrical panels for consumption by one or more usage appliances within the plurality of individual units; a plurality of usage current readers, each of the plurality of usage current readers being configured to track an amount of electrical energy used by an individual unit, each of the plurality of usage current readers being communicatively coupled with the data communication network; and a virtual grid server communicatively coupled with the production current reader and the plurality of usage current readers via the data communication network, the virtual grid server being configured to: obtain energy use information for a plurality of units electrically connected to at least one shared energy producing asset, the energy use information corresponding to a time period having a plurality of time intervals; obtain energy production information for the energy producing asset, the energy production information corresponding to the time period and the plurality of time intervals; obtain energy pricing information for each unit in the plurality of units, the energy pricing information corresponding to the time period and the plurality of time intervals, each unit having a per unit per time interval price of energy, a first per unit per time interval price of energy for a first unit during a first time interval being not equal to a second per unit per time interval price of energy for a second unit during the first time interval; and for each time interval in the time period: identify a quantity of energy used by each unit during the time interval; identify a quantity of energy produced by the energy producing asset during the time interval; apply at least a portion of the quantity of energy produced during the time interval to each unit's respective energy used during the time interval; if there is an excess of energy produced during the time interval after application of energy produced to each unit, add the excess of energy produced during the time interval to a shared virtual bank for the time interval; if there is an excess of energy used by each unit during the time interval, apply at least a portion of energy from a shared virtual bank for any time interval to each unit's respective energy used during the time interval; after application of energy produced during the time interval and energy from a shared virtual bank for any time interval to each unit's respective energy used during the time interval, identify any excess quantity of energy used by each respective unit during the time interval; and calculate an energy use cost allocation for each respective unit based on: the quantity of energy produced by the shared energy producing asset applied to the respective unit during each time interval; the quantity of energy from one or more shared virtual banks applied to the respective unit during each time interval; the excess quantity of energy used by the respective unit during each time interval; and the energy pricing information for each time interval.

12. The system of claim 11, wherein the virtual grid server is further configured to, after applying at least a portion of the quantity of energy produced during the time interval to each unit's respective energy used during the time interval: determine an average price of energy during the time interval, the average price based on a per unit price of energy during the time interval for least a portion of the plurality of units; and update an index for each unit based on the quantity of energy applied to the respective unit for the time interval and the determined average price of energy during the time interval.

13. The system of claim 11, wherein the virtual grid server is further configured to, after applying

at least a portion of energy from the shared virtual bank for any time interval to each unit's respective energy used during the time interval; identify one or more shared virtual bank time intervals from which energy was applied to a respective unit's energy used during the time interval; identify an average price of the quantity of energy in each of the one or more shared virtual bank time intervals; and update an index for each unit based on the quantity of energy from each of the one or more shared virtual bank time intervals applied to the respective unit for the time interval and the average price of the quantity of energy in each of the one or more shared virtual bank time intervals.

14. The system of claim 11, wherein the energy use information is obtained from a utility company.

15. The system of claim 11, wherein the energy use information for a particular unit is obtained from an energy meter for the particular unit.

16. The system of claim 11, wherein: the energy use information for a particular unit is obtained from one or more usage current readers; and each usage current reader is configured to track an amount of electrical energy used.

17. The system of claim 11, wherein the energy production information is obtained from one or more production current readers electrically coupled with the energy producing asset and configured to track an amount of electrical energy produced by the energy producing asset.
